

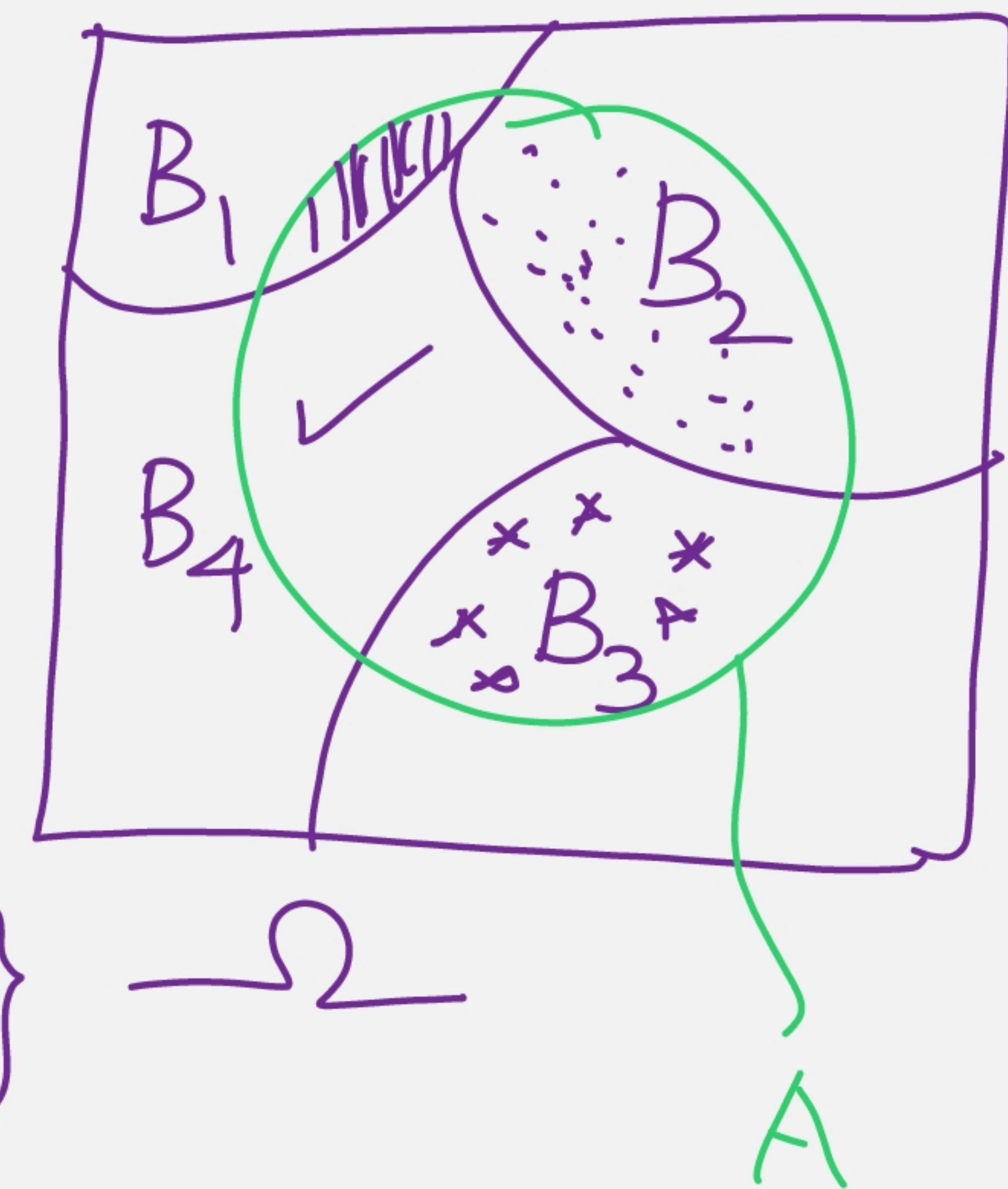
Bayes Theorem: Let $(\Omega, \mathcal{B}, P)$ be a probability space and A be an event with $P(A) > 0$. Let $B_1, B_2, \dots, B_n$ are pairwise disjoint set of events such that $\bigcup_{i=1}^n B_i = \Omega$. Then for $i = 1, 2, \dots, n$

$$P(B_i | A) = \frac{P(B_i) P(A | B_i)}{\sum_{r=1}^n P(B_r) P(A | B_r)}$$

Proof: $\Omega = \bigcup_{i=1}^n B_i$
 $= B_1 \cup B_2 \cup \dots \cup B_n$
 $\text{or, } \Omega \cap A = (B_1 \cup B_2 \cup \dots \cup B_n) \cap A$

$$\text{or, } A = (B_1 \cap A) \cup (B_2 \cap A) \cup \dots \cup (B_n \cap A).$$

$$\text{or, } P(A) = P\{(B_1 \cap A) \cup (B_2 \cap A) \cup \dots \cup (B_n \cap A)\} \rightarrow \Omega$$



$$\begin{aligned} P(B_i | A) &= \frac{P(B_i \cap A)}{P(A)} \\ &= \frac{P(B_i) P(A | B_i)}{P(A)} \\ &= \frac{P(B_i) P(A | B_i)}{\sum_{r=1}^n P(B_r) P(A | B_r)} \end{aligned}$$

Since $B_1, B_2, \dots, B_n$ are pairwise disjoint events, $A \cap B_1, A \cap B_2, \dots, A \cap B_n$ are pairwise disjoint sets.

$$\begin{aligned} \text{or, } P(A) &= P(B_1 \cap A) + P(B_2 \cap A) + \dots + P(B_n \cap A) \\ &= P(B_1) P(A | B_1) + P(B_2) P(A | B_2) + \dots + P(B_n) P(A | B_n) \end{aligned}$$

$$\text{or, } P(A) = \sum_{r=1}^n P(B_r) P(A | B_r)$$

Ex: Suppose a calculator manufacturer purchases his IC's from suppliers B_1, B_2, B_3 with 40% from B_1 , 30% from B_2 & 30% from B_3 . Suppose 1% of supply from B_1 is defective, 5% from B_2 & 10% from B_3 is defective. (a) What is the probability that a randomly selected IC from manufacturer's stock is defective?

(b) Suppose a randomly selected IC is found to be defective. What is the prob that it was supplied by supplier B_1 ?

Sol: $A \rightarrow$ "the IC is defective"

$$P(A) = \sum_{j=1}^3 P(A|B_j) P(B_j)$$

$$= P(A|B_1) P(B_1) + P(A|B_2) P(B_2) + P(A|B_3) P(B_3)$$

$$= 0.01 \times 0.4 + 0.05 \times 0.3 + 0.1 \times 0.3$$

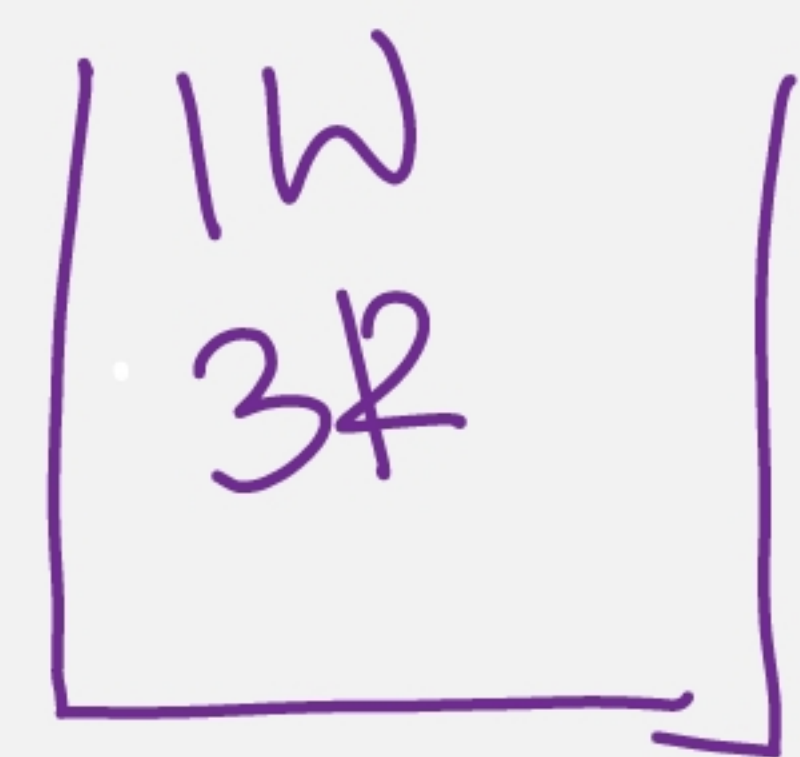
$$= 0.004 + 0.015 + 0.03$$

$$= 0.049.$$

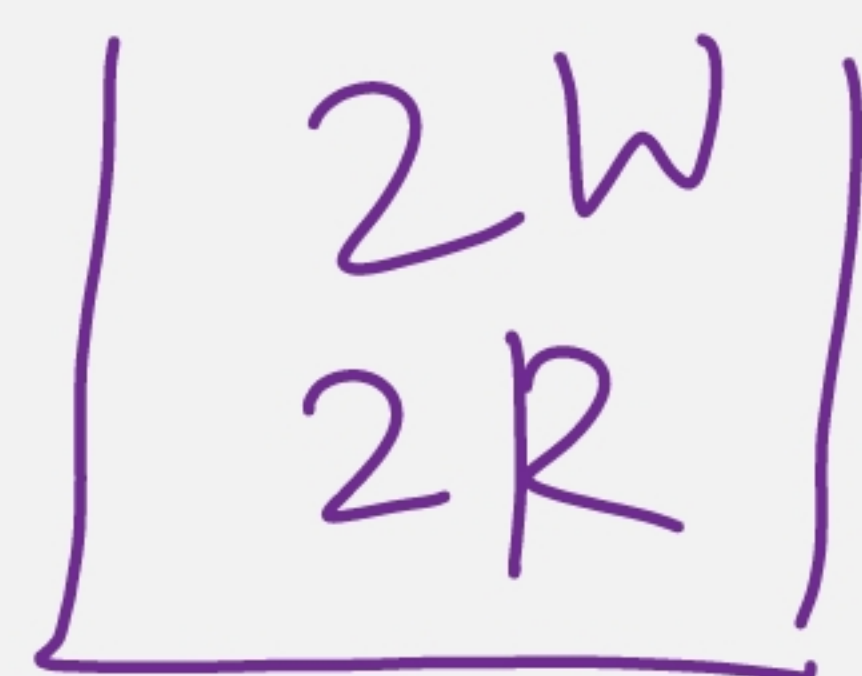
$$P(B_1|A) = \frac{P(B_1) P(A|B_1)}{P(A)}$$

$$= \frac{0.01 \times 0.4}{0.049} = \frac{4}{49}$$

$$P(B_3|A) = \frac{P(B_3) P(A|B_3)}{P(A)}$$



B_1



B_2



B_3

$B_i \rightarrow$ "ith box is chosen"

1, 2, 3 choose Box 1

4, 5 $\rightarrow$ choose Box 2

6 $\rightarrow$ choosing Box 3.

$P(B_2|R)$

$P(B_1|R)$

$P(B_3|W)$

W: white ball drawn

$$P(B_2|W) = \frac{P(B_2)P(W|B_2)}{P(B_1)P(W|B_1) + P(B_2)P(W|B_2) + P(B_3)P(W|B_3)}$$

$$= \frac{\frac{2}{6} \times \frac{2}{4}}{\frac{3}{6} \times \frac{1}{4} + \frac{2}{6} \times \frac{2}{4} + \frac{1}{6} \times \frac{3}{4}}$$

Tossing of two fair coins

$$\Omega = \{HH, HT, TH, TT\}$$

$A \rightarrow$ "Head on first coin"

$B \rightarrow$ "Head on second coin"

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} = P(A)$$

If $P(A|B) = P(A)$, then A and B are said to be independent events.

$\Rightarrow$

$$P(A|B) = P(A)$$

$$\text{or, } \frac{P(A \cap B)}{P(B)} = P(A)$$

$$\text{or, } \boxed{P(A \cap B) = P(A)P(B)}$$

Q. If A, B are independent events, prove that

(i) A and B^c are independent events.

(ii) A^c and B are independent events.

(iii) A^c and B^c are independent events.

$$m_3 = m_3' - 3m_2'm_1' + 2m_1'^3$$

$$\Rightarrow \boxed{m_3'} = m_3 + 3m_2'm_1' - 2m_1'^3$$

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$$\gamma_1 = \frac{m_3}{m_2^2}, \quad \beta_2 = \frac{m_4}{m_2}$$

$$\gamma_1 = +1, \quad \gamma_2 = \beta_2 = 3$$

$$\gamma_1, \gamma_2$$